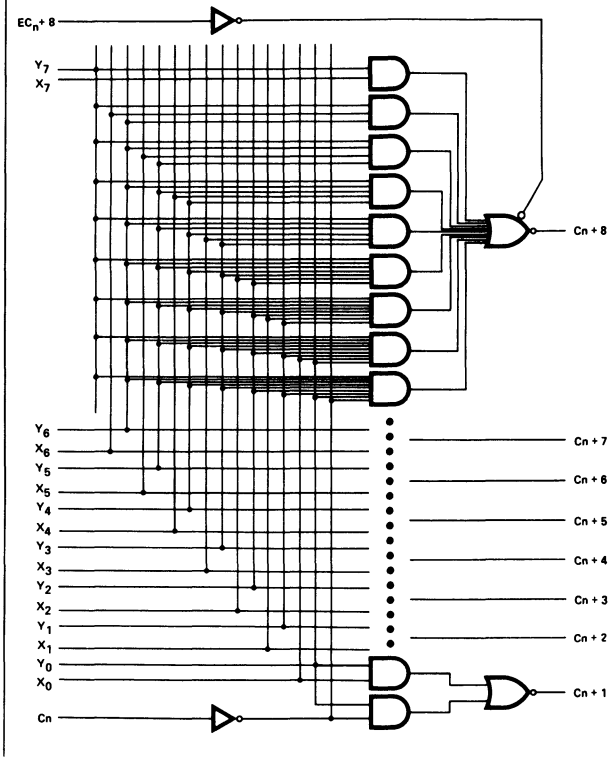
PIN DESCRIPTION

PIN	SYMBOL	NAME AND FUNCTION	TYPE
1,7,8,11 18,21,23 27	Y_0-Y_7	Standard carry look-ahead inputs	Active HIGH
2,5,6,10 19,20,24 26	X_0-X_7	Standard carry look-ahead inputs	Active HIGH
17	C_n	Carry input	Active LOW
4,9,12 13,15,16	C_{n+1}- C_{n+8}	Carry outputs	Active LOW
3	EC_{n+8}	C_{n+8} carry output enable	Active HIGH
28	V_{CC}	+5 volt supply	
14	GND	Ground	

LOGIC DIAGRAM



3003 LOGIC EQUATIONS

The 3003 Look-Ahead Generator is implemented in a compatible form for direct connection to the 3001 MCU and 3002 CPE. Logic equations for the 3003 are:

$$\overline{C_{n+1}} = Y_0X_0 + Y_0\overline{C}_n$$

$$\overline{C_{n+2}} = Y_1X_1 + Y_1Y_0X_0 + Y_1Y_0\overline{C}_n$$

$$\overline{C_{n+3}} = Y_2X_2 + Y_2Y_1X_1 + Y_2Y_1Y_0X_0 + Y_2Y_1Y_0\overline{C}_n$$

$$\overline{C_{n+4}} = Y_3X_3 + Y_3Y_2X_2 + Y_3Y_2Y_1X_1 + Y_3Y_2Y_1Y_0X_0 + Y_3Y_2Y_1Y_0\overline{C}_n$$

$$\overline{C_{n+5}} = Y_4X_4 + Y_4Y_3X_3 + Y_4Y_3Y_2X_2 + Y_4Y_3Y_2Y_1X_1 + Y_4Y_3Y_2Y_1Y_0X_0 + Y_4Y_3Y_2Y_1Y_0\overline{C}_n$$

$$\overline{C_{n+6}} = Y_5X_5 + Y_5Y_4X_4 + Y_5Y_4Y_3X_3 + Y_5Y_4Y_3Y_2X_2 + Y_5Y_4Y_3Y_2Y_1X_1 + Y_5Y_4Y_3Y_2Y_1Y_0X_0 + Y_5Y_4Y_3Y_2Y_1Y_0\overline{C}_n$$

$$\overline{C_{n+7}} = Y_6X_6 + Y_6Y_5X_5 + Y_6Y_5Y_4X_4 + Y_6Y_5Y_4Y_3X_3 + Y_6Y_5Y_4Y_3Y_2X_2 + Y_6Y_5Y_4Y_3Y_2Y_1X_1 + Y_6Y_5Y_4Y_3Y_2Y_1Y_0X_0 + Y_6Y_5Y_4Y_3Y_2Y_1Y_0\overline{C}_n$$

$\overline{C_{n+8}}$ = High Impedance State when $EC_n + 8$ Low

$$\overline{C_{n+8}} = Y_7X_7 + Y_7Y_6X_6 + Y_7Y_6Y_5X_5 + Y_7Y_6Y_5Y_4X_4 + Y_7Y_6Y_5Y_4Y_3X_3 + Y_7Y_6Y_5Y_4Y_3Y_2X_2 + Y_7Y_6Y_5Y_4Y_3Y_2Y_1X_1 + Y_7Y_6Y_5Y_4Y_3Y_2Y_1Y_0X_0 + Y_7Y_6Y_5Y_4Y_3Y_2Y_1Y_0\overline{C}_n \text{ when } EC_n + 8 \text{ high}$$